Practical 3.1

```
import pandas as pd

# 1. Load the Adult dataset
df = pd.read_csv('adult (3.1).csv')

# 2. Group by categorical variable ('income') and calculate stats for 'age'
# We calculate: mean, median, minimum, maximum, and standard deviation
summary_stats = df.groupby('income')['age'].agg(['mean', 'median', 'min', 'max', 'std'])

print("--- Summary Statistics of Age Grouped by Income ---")
print(summary_stats)

# 3. Create a list that contains a numeric value for each response
# Here, we create a list of the average age for each income category
income_categories = summary_stats.index.tolist()
mean_age_list = summary_stats['mean'].tolist()

print("\nIncome Categories:", income_categories)
print("Mean Age List:", mean_age_list)
```

Practical 3.2

```
import pandas as pd

# 1. Load the Iris dataset
df = pd.read_csv('Iris Dataset Prac (1,3.2,6,10).csv')

# 2. Use describe() grouped by Species
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```
# This automatically gives mean, std, min, max, and percentiles (25%, 50%, 75%)
species_stats = df.groupby('Species').describe()

# 3. Display details for a specific column (e.g., 'SepalLengthCm')
# to keep the output readable
print("--- Statistical Details for SepalLengthCm by Species ---")
print(species_stats['SepalLengthCm'])

print("\n--- Statistical Details for PetalLengthCm by Species ---")
print(species_stats['PetalLengthCm'])
```